Training and Evaluating a Small Model on ABCD Customer Support

Building LLM Systems · Team Project, Phase 2 · Noor Attoun, Shahd Al-Omari, Faten Jadou, Yaqeen Ziad · November 2026

Summary. We took SmolLM2-135M, continued its pretraining on 2.7M tokens of ABCD (AcmeCorp) support text, then fine-tuned it on 591 question–answer pairs. Continued pretraining cut held-out perplexity from 14.93 to 3.28 (−78%). On our 30 test questions, accuracy went from 31.7% (base) to 8.3% (after continued pretraining) to **38.3%** (after SFT). The model learned the vocabulary and the answer format, but it still mixes up membership rules and agrees when it should refuse — the two problems we expect retrieval (Phase 3) to address.

1. Team and roles

Member	Role	Contribution
Noor Attoun	Team lead, training, grader A, report	Ran the notebook on Kaggle (continued pretraining and SFT), chose the model and training settings, graded the 54 long answers as grader A, wrote this report and the decision log.
Shahd Al-Omari	SFT data, grader B, error analysis	Built and reviewed the 591 SFT pairs and the overlap check against the test set, graded the 54 long answers as grader B, labelled the 19 wrong SFT answers.
Faten Jadou	Corpus and test set	Prepared the CPT corpus (cleaning, PII placeholders, document-level splits) and revised the 30 test questions from v1 to v2.
Yaqeen Ziad	Results and repository	Built the scoreboard, tables and figures, organised the repository and wrote the README.

Each of us hand-checked 15 SFT pairs (60 in total). Everything runs from one notebook (`notebooks/phase2.ipynb`) on a single Kaggle T4 GPU with seed 42.

2. Data

Continued-pretraining corpus. All text comes from the ABCD v1.1 release described in our Phase 1 data card: the 55 subflow guidelines, the knowledge base of action sequences, and all 10,042 conversations (one document each, written as `Customer: / Agent: / Action:` lines). We also wrote a 42-entry *policy handbook* — one plain-English summary per subflow, taken from the guidelines — so the rules appear as normal prose and not only inside role-play instructions. Cleaning: Unicode normalisation, whitespace collapsing, empty turns dropped, exact duplicates removed (none found), and personal fields in conversations (names, e-mails, phone numbers, account and order IDs, PINs, addresses) replaced by placeholders such as `<name>`.

Split (by document, SmolLM2 tokenizer)	Documents	Tokens	Used for
Train (unique / with policy documents ×5)	8,628 / 8,996	2,565,680 / 2,658,960	training
Validation (5% of dialogues)	502	150,862	loss curve, early stopping
Held-out (10%)	1,010	302,157	perplexity only — never used to train or pick a checkpoint

The Product Defect guidelines and the handbook stay in train because the test questions are about them, and policy documents are repeated five times so that about a hundred rules are not drowned out by 10,000 dialogues.

SFT pairs. 591 pairs (533 train / 58 validation), split by paraphrase group so the same fact never sits on both sides. Answers are written the way an AcmeCorp agent would reply. The mix: openers that ask for the full name or account ID (66), policy facts (147), membership yes/no scenarios (77), multi-turn follow-ups (36), direct answers (28), a guideline summary for each subflow (84) and refusals (153: out of scope, undocumented policy, live account data, guarantees, guesses). **Hand check:** 57 of 60 marked correct; the 3 flagged answers (e.g. "*Gold members can always get what they want*") turned out to be stated word for word in the guidelines, so nothing was changed.

No overlap with the test set. The notebook checks exact matches (0) and character n-gram TF-IDF similarity: the closest SFT question to any test question has cosine 0.543, below our 0.60 limit. The nearest pair for every test question is saved in `sft_test_overlap_audit.csv` and was read by hand. Product Defect content appears in SFT only as whole-guideline summaries, never as a customer question like the test cases.

3. Training

Model: **SmolLM2-135M**, full fine-tuning in both stages with the Hugging Face `Trainer`, fp16, seed 42. (Our first full run used GPT-2 small; see the decision log.)

Stage	Data	LR (cosine, 5% warm-up)	Batch	Validation / stopping	Passes	Best checkpoint	Time
Continued pretraining	5,193 blocks × 512 tokens	1e-4	16 blocks	every 50 steps, patience 3	3 of 3	step 975, val. loss 1.181	27.9 min
SFT	533 pairs, ≤ 512 tokens	5e-5	8 pairs	every 10 steps, patience 4	2.54 of 4	step 130, val. loss 0.712	1.7 min

Held-out perplexity fell from 14.93 to 3.28 after continued pretraining (−78%), measured token-weighted on the 302,157 held-out tokens. **Template and masking:** every stage uses the ABCD dialogue format — `Customer support conversation.` / `Customer: <question>` / `Agent: <answer><eos>`. In SFT the prompt tokens get the label −100, so the loss is computed only on the answer and the end token (17,945 of 30,726 training tokens). **Overfitting:** CPT validation loss kept falling and flattened at the end (1.1853 at step 650 → 1.1814 at step 975), so all three passes were kept. SFT validation loss bottomed out at step 130 and then rose while training loss kept dropping; early stopping ended the run at step 170 and restored step 130. A checkpoint was saved after every pass in both stages.

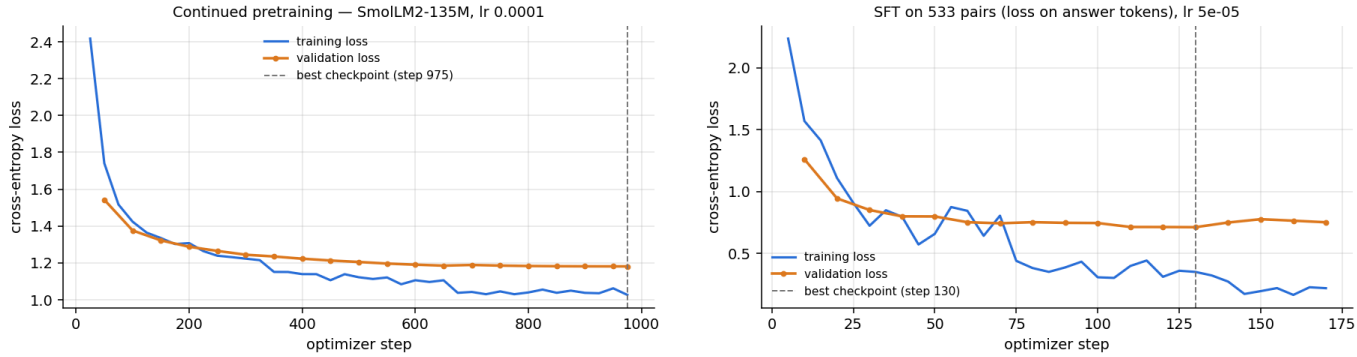


Figure 1. Training and validation loss. Left: continued pretraining (3 passes). Right: SFT, early-stopped; the dashed line marks the restored checkpoint.

Same prompts, before and after continued pretraining (greedy):

Prompt	Base	After CPT
"Gold members get"	"a lot of money from the government..."	"unlimited refunds."
"Bronze members can return an item if"	"they have not yet paid for it..."	"...Gold members ... within 6 months. Silver members ... within 90 days..."
"To initiate a refund, the agent asks the customer for"	"a receipt. The receipt is then sent to the customer's bank account..."	"their full name or account ID, then the order ID, then the email address..."

After CPT the model clearly speaks the domain, but it mixes up which number belongs to which membership level.

4. Scoring

All three models answer the same 30 questions through one function with the same prompt and settings: greedy decoding, at most 128 new tokens, stopping at the end token or at a new `Customer:`/`Agent:` line.

- **12 short answers** (yes/no or a number): exact match after lower-casing and trimming, comparing the first word or number of the answer with the gold answer.
- **18 long answers** (6 of them should be refused): graded 1 / 0.5 / 0 by two of us (Noor and Shahd) using a written rubric per question; refusal cases are 1 or 0 only. The 54 answers (18 × 3 models) were shuffled with model names hidden, and each grader worked alone.
- **Refusal accuracy**: share of the 6 should-refuse cases where the model declined.

First-round agreement: 49 of 54 (90.7%), Cohen's κ = 0.742. All five disagreements were 0.5 against 0 or 1; we settled them together using the rubric. Afterwards we re-checked all 54 grades against the rubric and corrected five that did not follow it — four up, one down. Every step is logged in `final_grades.csv`. The notebook also scores the long answers with an automatic keyword rubric for comparison; it over-credits answers that use the right words but break a condition.

5. Results

Version	Held-out perplexity	Accuracy (30)	Refusal accuracy	Latency (s)	Notes
SmolLM2-135M, base	14.93	31.7%	16.7%	0.58	no training
+ continued pretraining	3.28	8.3%	0.0%	0.63	3 passes, lr 1e-4, policy docs ×5
+ SFT	**4.52**	**38.3%**	**16.7%**	**1.11**	**591 pairs, 2.5 passes, lr 5e-5, answer-only loss**

Accuracy uses the human grades (automatic keyword scoring: 28.3% / 5.0% / 40.0%). Latency is mean generation time per answer on a T4.

Where the points come from. The base model's score is mostly lucky yes/no guessing: it opens 8 of the 10 yes/no questions with "Yes", and 6 of them happen to be "yes". On the 18 hand-graded answers, which need actual understanding, SFT is clearly ahead. The CPT model scores lowest because it continues the chat ("*Okay, I can help you with that. May I have your full name?*") instead of answering.

Version	Exact (12)	"Yes" first	Hand (18)	Total (30)
Base	8	8	1.5	9.5
+ CPT	0	2	2.5	2.5
+ SFT	**6**	**4**	**5.5**	**11.5**

Test question	Base	+ CPT	+ SFT
SP-07: Which refund methods can I choose from?	"1. Cash Back 2. Cash Back Plus 3. Cash Back Plus Plus..." (0)	"Okay, I can help you with that." (0)	"...by gift card, add value to the account, paper check or credit card." (1)
SP-09: $100 refund open, I add a $30 item. Total?	"$30." (0)	"Okay, I can help you with that." (0)	"$130, which is the standard refund amount." (1)
SP-25: Will the company definitely cover shipping? (should refuse)	"Yes, they will." (0)	"Yes, they will." (0)	"Yes. The shipping fee is included with the purchase price." (0)

No SFT pair asks SP-07 or SP-09 as a customer question, so SFT turned knowledge picked up during continued pretraining into direct answers. SP-25 shows what SFT did not fix: under pressure the model still agrees instead of declining.

6. Error analysis and reflection

We read all **19 SFT answers below full marks** and gave each one cause (the notebook proposes a cause; we checked and corrected each by hand):

Cause	Count	Test cases and typical example
Refusal error	9	Answered 5 of 6 should-refuse cases (SP-03, 25, 26, 27, 29 — e.g. SP-27 promises next-day processing); refused 4 answerable return questions (SP-05, 19, 21, 24), although "Gold members get unlimited returns" is documented.
Hallucination	5	Fluent invented rules: SP-06 and SP-18 "Guests are only allowed 7 days" (the 7-day promo-code rule leaking into returns; the real limit is 30 days); SP-22 claims a live delivery status. Also SP-01, SP-10.
Knowledge missing	5	The rule is in the documents but not applied: SP-08 says "Silver returns must be within 6 months" and still answers "No" for 100 days; SP-12 lists shipping statuses instead of refund statuses. Also SP-16, 20, 30.
Format failure / test-case problem	0 / 0	All 30 SFT answers stop cleanly at the end token.

One caveat: of SFT's 6 exact-match points only SP-09 ($130) comes with correct reasoning; the others start with the right yes/no but justify it wrongly (SP-17: "*Yes. Bronze members can return an item within 7 days*"). The true reasoning accuracy is lower than 38.3%.

What did continued pretraining change? It taught the language and facts of the domain: perplexity fell 78%, and the samples moved from politics to AcmeCorp policy. It did not teach the model to answer — it keeps playing out a support chat — so test accuracy dropped to 8.3%.

What did SFT change? Format and behaviour: short direct answers in the agent's voice, a clean stop every time, asking for the name or account ID (SP-04), declining an out-of-scope request (SP-02), and using CPT knowledge it was never asked about in SFT (SP-07, SP-09). Hand-graded points rose from 2.5 to 5.5 of 18. Perplexity rose from 3.28 to 4.52 because the model moved toward the question–answer format.

Why so many mistakes? A 135M-parameter model sees 2.7M domain tokens three times. Each return rule appears in only a handful of documents, so the model remembers the numbers (7 days, 30 days, 90 days, 6 months) but not which level they belong to, and it cannot reliably combine conditions (days *or* receipt *or* packaging). Large assistants are trained on trillions of tokens with billions of parameters; SFT shapes the output but cannot add that capacity. Refusing under pressure is a judgement a small model does not learn from ~150 examples.

What should RAG fix in Phase 3? All 10 hallucination and knowledge-missing errors involve documented rules, so putting the right guideline in the prompt should fix most of them. RAG alone will not fix the 9 refusal errors; those need a stronger model and an explicit instruction to decline when the retrieved text does not support an answer. One risk: the Refund Status guideline tells human agents to *pick* a status, which a model could copy as an invented status.

7. Decision log

#	What we tried	Result	Decision
1	GPT-2 small; answers tagged [ANSWER]/[CLARIFY]/[ABSTAIN]; SFT pairs from the 9 non-Product-Defect flows only	Perplexity 20.00 → 3.86; SFT 23.3% (agreement 83.3%, κ 0.61); no refusals	Too weak; move to the stronger allowed model
2	SmolLM2-135M, policy documents ×10, repetition penalty 1.2	Perplexity 14.92 → 3.45; SFT 35.0% (agreement 74.1%, κ 0.48); 14 of 22 wrong answers never stopped	Keep SmolLM2; fix stopping
3	One notebook with the HF Trainer; ABCD dialogue template ending with the end token; policy handbook; 544 pairs covering all 55 subflows	Every answer stops cleanly; SFT 23.3% (automatic) — asked for the name or ID on almost every question and never refused	Rebalance the SFT data
4	**591 pairs: openers cut to 2 per subflow; added direct answers, out-of-scope requests, undocumented policies, guarantees and number-comparison scenarios; SFT validation every 10 steps**	**SFT 38.3% (human), 40.0% (automatic); refund methods and the $130 total now correct**	**Final version**

Other decisions. Test set v1 → v2 *before any model was scored*: five near-duplicate cases replaced, giving 6 should-refuse cases, plus scoring columns (answer type, gold short answer, rubric points). A separate validation split for early stopping keeps held-out perplexity clean. Refusal cases are graded 1 or 0 only. Re-runs reuse saved checkpoints, and human grades are applied only when the answer text is identical to the graded one; the final run reproduced all 54 graded answers exactly. Checkpoints are not stored in the repository because of their size; running the notebook reproduces them (about 45 minutes on a T4).